Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A light of wavelength λ and intensity I is incident normally on the surface. If reflection coefficient of surface is r, the pressure exerted by light on the surface is equal to

(A) $\frac{2I}{c}$

(B) $\frac{I}{c}r$

(C) $\frac{I}{c}(1+r)$

(D) $\frac{I}{c}(1-r)$

Q.2

A proton has kinetic energy $E = 100\text{keV}$ which is equal to that of a photon. The wavelength of photon is λ_2 and that of proton is λ_1. The ration of λ_2/λ_1 is proportional to

(A) E^2

(B) $E^{1/2}$

(C) E^{-1}

(D) $E^{-1/2}$

Q.3

An electron is accelerated through a potential difference of V volt. It has a wavelength λ associated with it. Through what potential difference an electron must be accelerated so that its de Broglie wavelength is the same as that of a proton? Take mass of proton to be 1837 times larger than the mass of electron.

(A) V volt

(B) $1837\ \text{V}$ volt

(C) $\text{V}/1837$ volt

(D) $\sqrt{1837}\ \text{V}$ volt

Q.4

Assume that the de Broglie wave associated with an electron can form a standing wave between the atoms arranged in a one dimensional array with nodes at each of the atomic sites. It is found that one such

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standing wave is formed if the distance d between the atoms of the array is $2\tilde{A} \ldots$. A similar standing wave is again formed if d is increased to $2.5\tilde{A} \ldots$ but not for any intermediate value of d. The energy of the electrons in electron volt and the least value of d for which the standing wave of the type described above can form, respectively are

(A) $112\mathrm{eV}, 1.5\ \mathrm{A}^{\circ}$

(B) $151\mathrm{eV}, 0.5\ \mathrm{A}^{\circ}$

(C) $211\mathrm{eV}, 1.5\ \mathrm{A}^{\circ}$

(D) None of these

Q.5

When a beam of $10.6\mathrm{eV}$ photons of intensity $2.0\ \mathrm{W/m^2}$ falls on a platinum surface of area $1.0 \times 10^{-4}\ \mathrm{m}^2$ and work function $5.6\mathrm{eV}, 0.53\%$ of the incident photons eject photoelectrons, then the number of photoelectrons emitted per second and their minimum \& maximum energies (in eV) $\left[\right.$ Take $1\mathrm{eV} = 1.6 \times 10^{-19}\ \mathrm{J}\left.\right]$ are respectively

(A) $1.18 \times 10^{10}, 2\mathrm{eV}, 5\mathrm{eV}$

(B) $1.18 \times 10^{14}, 0\mathrm{eV}, 5\mathrm{eV}$

(C) $2.18 \times 10^{13}, 0\mathrm{eV}, 5\mathrm{eV}$

(D) $3.11 \times 10^{11}, 1\mathrm{eV}, 6\mathrm{eV}$

Q.6

When a surface is irradiated with light of wavelength $4950\tilde{A}$, a photocurrent appears which vanishes if a retarding potential greater than $0.6\ \mathrm{V}$ is applied across the phototube. When a different source of light is used, it is found that the critical retarding potential is changed to 1.1
V. Find the wavelength of second source.

(A) $4077\tilde{A}$

(B) $992\tilde{A}$

(C) $628\tilde{A}$

(D) $238\tilde{A}$

Q.7

Find the frequency of light which ejects electrons from a metal surface fully stopped by a retarding potential of $3\ \mathrm{V}$. The photoelectric effect begin in this metal at frequency of 6×10^{14} per second.

(A) $1.32 \times 10^{15}\ \mathrm{Hz}$

(B) $3.28 \times 10^{14}\ \mathrm{Hz}$

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(C) 6.22×10^{15} Hz

(D) 2.22×10^{11} Hz

Q.8

The potential energy of a particle of mass m_{15} given by $V(x) = \left\{ \begin{array}{cc} E_0; & 0 \leq x \leq 1 \\ 0; & x > 1 \end{array} \right\}$

λ_1 and λ_2 are the de-Broglie wavelengths of the particle, when $0 \leq x \leq 1$ and $x > 1$ respectively. If the total energy of particle is $2E_0$, find λ_1/λ_2

(A) $\sqrt{2}$

(B) $\sqrt{3}$

(C) $\sqrt{5}$

(D) $2\sqrt{2}$

Q.9

Light of wavelength 180 nm ejects photoelectron from a plate of a metal whose work function is 2 eV. If a uniform magnetic field of 5×10^{-5} T is applied parallel to plate, what would be the radius of the path followed by electrons ejected normally from the plate with maximum energy?

(A) 1.239 m

(B) 0.149 m

(C) 3.182 m

(D) 2.33 m

Q.10

A cylindrical rod of some laser material 5×10^{-2} m long and 10^{-2} m in diameter contains 2×10^{25} ions per m^3. If on excitation all the ions are in the upper energy level and de-excite simultaneously emitting photons in the same direction, calculate the maximum energy contained in a pulse of radiation of wavelength 6.6×10^{-7} m. If the pulse lasts for 10^{-7} s. the average power of the laser during the pulse is

(A) $532\mathrm{MW}$

(B) $352\mathrm{MW}$

(C) $235\mathrm{MW}$

(D) $325\mathrm{MW}$

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Q.11

Radiation of wavelength λ, is incident on a photocell. The fastest emitted electron has speed v. If the wavelength is changed to $\frac{3\lambda}{4}$, the speed of the fastest emitted electron will be:

(A) $= v\left(\frac{4}{3}\right)^{\frac{1}{2}}$

(B) $= \mathrm{v}\left(\frac{3}{4}\right)^{\frac{1}{2}}$

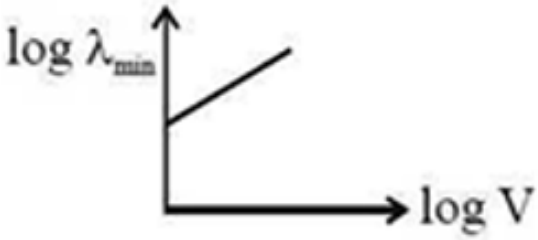
(C) $> v\left(\frac{4}{3}\right)^{\frac{1}{2}}$

(D) $< v\left(\frac{4}{3}\right)^{\frac{1}{2}}$

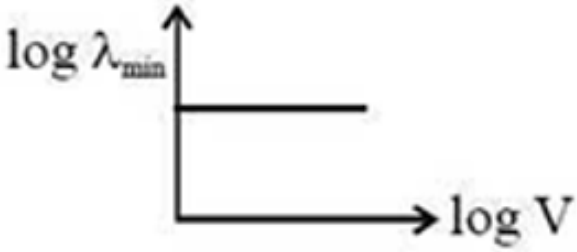
Q.12

An electron beam is accelerated by a potential difference V to hit a metallic target to produce $\mathrm{X}-$ rays. It produces continuous as well as characteristic X-rays.If $\lambda_{\min}$ is the smallest possible wavelength of X-ray in the spectrum, the variation of $\log \lambda_{\min}$ with $\log \mathrm{V}$ is correctly represented in :

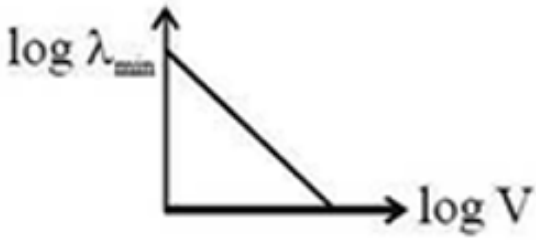
(A)



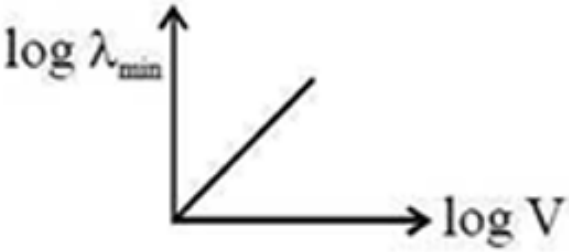
(B)



(C)



(D)



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Q.13

A point source of light is placed at the centre of curvature of a hemispherical surface. The radius of curvature is R and the inner surface is completcly reflecting The force on the hemisphere due to the light falling on it if the source emits a power $\mathrm{P}[\mathrm{c}$ is the speed of light $]$

(A) $\frac{2P}{C}$

(B) $\frac{P}{2C}$

(C) $\frac{5\mathrm{P}}{2\mathrm{C}}$

(D) $\frac{4P}{3C}$

Q.14

A photon of frequency f causes the emission of a photoelectron of maximum kinetic energy E_k from a metal. If a photon of frequency $3f$ is incident on the same metal, the maximum kinetic energy of the emitted photoelectron

(A) equals $3E_k$

(B) is greater than $3E_k$

(C) is less than $3E_k$

(D) may be equal to, less than or, greater than $3E_k$

Q.15

A 5 watt source emits monochromatic light of wavelength $5000\tilde{A}\ldots$. When placed $0.5\ \mathrm{m}$ away, it liberates photoelectrons from a photosensitive metallic surface. When the source is moved to a distance of $1.0\ \mathrm{m}$, the number of photoelectrons liberated will be reduced by a factor of

(A) 8

(B) 16

(C) 2

(D) 4

Q.16

The work functions of metals A and B are in the ratio $1:2$. If light of frequencies f and 2 f are incident on the surfaces of A and B respectively, the ratio of the maximum kinetic energies of photoelectrons emitted is $(\mathrm{f}$ is greater than threshold frequency of $\mathrm{A}, 2\mathrm{f}$ is greater than threshold frequency of B)

(A) $1:1$

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(B) $1:2$

(C) $1:3$

(D) $1:4$

Q.17

The fastest photoelectrons emitted from a metallic surface during the photoelectric effect have an energy of 8eV. The same photons when incident on atomic hydrogen, are strongly absorbed, exciting the atoms from the ground state to the first excited state. The work function of the metal is

(A) 5.6eV

(B) 2.2eV

(C) 4.6eV

(D) 11.4eV

Q.18

An ultraviolet light bulb, emitting $400\ \text{nm}$ and an infrared light bulb, emitting at $700\ \text{nm}$, each are rated at $130\ \text{W}$. Then the ratio of the number of photons emitted per second by the UV and IR sources is —

(A) 0.57

(B) 1.75

(C) 28

(D) 0.04

Q.19

A source of light is placed at a distance of $50\ \text{cm}$ from a photocell and the stopping potential is found to be V_0. If the distance between the light source and photocell is made $25\ \text{cm}$, the new stopping potential will be

(A) $2\ \text{V}_0$

(B) $\text{V}_0/2$

(C) V_0

(D) $4\ \text{V}_0$

Q.20

A sensor is exposed for time t to a lamp of power P placed at a distance l. The sensor has an opening that

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is $4\ \mathrm{d}$ in diameter. Assuming all energy of the lamp is given off as light, the number of photons entering the sensor if the wavelength of light is λ is

(A) $\mathrm{N} = \mathrm{P}\lambda \mathrm{d}^2 \mathrm{t}/\mathrm{hc}l^2$

(B) $\mathrm{N} = 4\mathrm{P}\lambda \mathrm{d}^2 \mathrm{t}/\mathrm{hc}l^2$

(C) $\mathrm{N} = \mathrm{P}\lambda \mathrm{d}^2 \mathrm{t}/4\mathrm{hc}l^2$

(D) $\mathrm{N} = \mathrm{P}\lambda \mathrm{d}^2 \mathrm{t}/16\mathrm{hc}l^2$

Answers & Solutions

Q.1 Answer:

$\frac{I}{c}(1+r)$

Solution:

$F \quad F = \frac{dP}{dt} = \left(\frac{IA}{hc/\lambda}\right)(1+r)\frac{h}{\lambda}$
$\Rightarrow P = \frac{F}{A} = \frac{I}{c}(1+r)$

Q.2 Answer:

$E^{-1/2}$

Solution:

For photon, $\lambda_2 = \frac{hc}{E}$
For proton, $p = \sqrt{2mE}\ \lambda_1 = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$
$\frac{\lambda_2}{\lambda_1} = \frac{hc}{E \times \frac{h}{\sqrt{2mE}}} \propto E^{-1/2}$

Q.3 Answer:

$\mathrm{V}/1837$ volt

Solution:

$\lambda = \frac{\mathrm{h}}{\sqrt{2\mathrm{mqV}}}$ **and** $\mathrm{mV} =$ **constant**
$1837\ \mathrm{V}' = 1 \times \mathrm{V}$ **or** $\mathrm{V}' = \frac{\mathrm{V}}{1837}$ **volt**

Q.4 Answer:

$151\mathrm{eV}, 0.5\ \mathrm{A}^\circ$

Solution:

As nodes are formed at each of the atomic sites, hence $2\mathring{A} = n(\lambda/2) \rightarrow (1)$ **and** $2.5\mathring{A}A = (\mathrm{n}+1)\lambda/2$
and $2.5\ \mathrm{A} = (\mathrm{n}+1)\lambda/2 \therefore 2.5/2 = \mathrm{nt1/n}.5/4 = \mathrm{n}+1/\mathrm{n}$ **or** $\mathrm{n} = 4$
Hence from equation (1)

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$2\ 2^{0}\mathrm{A} = 4\lambda/2$ **i.e.** $\lambda = 1\mathring{A}^{0}$

Now, de Brogile wavelength is given by,

$$\lambda = \mathrm{h}/\sqrt{2}\mathrm{mk} \text{ or } \mathrm{k} = \mathrm{h}^2/\lambda^2 = 2\ \mathrm{m}$$
$$\therefore \mathrm{k} = \left(6.63 \times 10^{-34}\right)^2 / \left(1 \times 10^{-10}\right)^2 \times 2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19}\mathrm{eV}$$
$$\mathrm{k} = 151\mathrm{eV}$$

θ will be minimum, when
$\mathrm{n} = 1,\ \mathrm{d}_{\min} = \lambda/2 = 1\ \mathrm{A}/2 = 0.5\ \mathrm{A}$

Q.5 Answer:

$1.18 \times 10^{14}, 0\mathrm{eV}, 5\mathrm{eV}$

Solution:

Given,

Intensity, $\mathrm{I} = 2\mathrm{Wm}^2$

Area, $\mathrm{A} = 10^{-4}\ \mathrm{m}^2$

Energy incident on the surface per second,

$\mathrm{P} = \mathrm{IA}$

$= 2.0 \times 1.0 \times 10^{-4}$

$= 2 \times 10^{-4}\ \mathrm{J\ s}^{-1}$

Energy of each photon,

$= 10.6\mathrm{eV}$

$= 10.6 \times 1.6 \times 10^{-19}\ \mathrm{J}$

$\therefore$ Number of photons incident on the surface

$= \frac{2 \times 10^{-4}}{10.6 \times 1.6 \times 10^{-19}}$

Number of photoelectrons emitted,

$= \frac{0.53}{100} \times \frac{2 \times 10^{-4}}{10.6 \times 1.6 \times 10^{-19}}$

$= 6.25 \times 10^{11}$

According to Einstein's photoelectric equation, maximum KE of photoelectrons,

$\mathrm{E}_{\mathrm{k}} = \epsilon - \mathrm{W} = 10.6\mathrm{eV} - 5.6\mathrm{eV}$

$= 5\mathrm{eV}$

Q.6 Answer:

$4077\tilde{A}$

Solution:

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By Einstein equation

$hf = W_0 + eV_0$

$W_0 = \frac{hc}{\lambda} - eV_0$

$= \frac{\left(6.63\times10^{-34}\right)\times\left(3\times10^{8}\right)}{4950\times10^{-10}} - \left(1.6 \times 10^{-19}\right) \times 0.6$

$= 3.12 \times 10^{-19}\ \mathrm{J}$

For the second surface

$hf' = W_0 + eV_0'$

or $\frac{hc}{\lambda'} = 3.12 \times 10^{-19} + 1.6 \times 10^{-19} \times 1.1$

$\therefore \quad \lambda' = 4077\tilde{A} \ldots$

Q.7 Answer:

$1.32 \times 10^{15}\ \mathrm{Hz}$

Solution:

The work function of the metal $W_0 = \mathrm{hf_0} = \left(6.63 \times 10^{-34}\right) \times \left(6 \times 10^{14}\right)$

$= 3.98 \times 10^{19}\ \mathrm{J}$

If f **is the frequency of the light, then** $hf = W_0 + eV_0$

$\therefore \quad f = \frac{W_0+eV_0}{h}$

$= \frac{3.98\times10^{-19}+1.6\times10^{-19}\times3}{6.63\times10^{-34}} = 1.32 \times 10^{15}\ \mathrm{Hz}$

Q.8 Answer:

$\sqrt{2}$

Solution:

Total energy, $\mathrm{E} = \mathrm{T} + \mathrm{V}$ **where** $\mathrm{T} =$ **Kinetic energy and** $\mathrm{V} =$ **potential energy.**

When $0 \leq x \leq 1, 2\mathrm{E_0} = \mathrm{T} + \mathrm{E_0} \Rightarrow \mathrm{T} = \mathrm{E_0}$

de-Broglie wavelength, $\lambda_1 = \frac{\mathrm{h}}{\sqrt{2\mathrm{mT}}} = \frac{\mathrm{h}}{\sqrt{2\mathrm{mE_0}}}$

When $\mathrm{x} > 1, 2\mathrm{E_0} = \mathrm{T} + 0$ **or** $\mathrm{T} = 2\mathrm{E_0}$

now, $\lambda_2 = \frac{\mathrm{h}}{\sqrt{2\mathrm{mT}}} = \frac{\mathrm{h}}{\sqrt{2\ \mathrm{m2E_0}}}$

Thus, $\frac{\lambda_1}{\lambda_2} = \sqrt{2}$

Q.9 Answer:

$0.149\ \mathrm{m}$

Solution:

If $\mathrm{v_{max}}$ **is the speed of the fastest electron emitted from the metal surface, then** $\frac{hc}{\lambda} = W_0 + \frac{1}{2}mv_{\max}^2$

$\frac{\left(6.63\times10^{-34}\right)\times\left(3\times10^{8}\right)}{\left(180\times10^{-9}\right)}$

$= 2 \times \left(1.6 \times 10^{-19}\right) + \frac{1}{2}\left(9.1 \times 10^{-31}\right) v_{\max}^2$

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$\therefore \quad v = 1.31 \times 10^6 \text{ m/s}$

The radius of the electron is given by $r = \frac{mv}{qB} = \frac{\left(9.1\times10^{-31}\right)\times\left(1.31\times10^{6}\right)}{(1.6\times10^{-19})\times(5\times10^{-9})} = 0.149 \text{ m}$

Q.10 Answer:

235MW

Solution:

Total number of ions in the rod $N = \left(2 \times 10^{25}\right) \times \left[\frac{\pi}{4} \times \left(10^{-2}\right)^2\right] \times \left(5 \times 10^{-2}\right)$

$= 7.85 \times 10^{19}$

The energy of excitation $E = N \times \frac{hc}{\lambda}$

$= \left(7.85 \times 10^{19}\right) \times \frac{\left(6.6\times10^{-34}\right)\times\left(3\times10^{8}\right)}{6.6\times10^{-7}}$

$= 23.55 \text{ J}$

Average power $P = \frac{E}{t} = \frac{23.55}{10^{-7}} = 23.55 \times 10^7 \text{ W} = 235\text{MW}$

Q.11 Answer:

$> v\left(\frac{4}{3}\right)^{\frac{1}{2}}$

Solution:

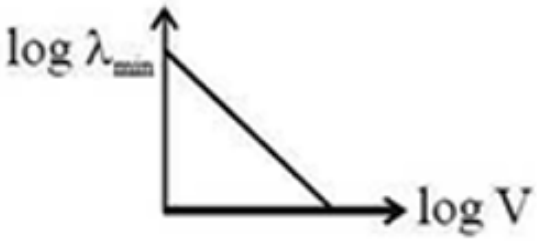
$\text{hv}_0^2 - \text{hv}_0 = \frac{1}{2}\text{mv}^2$

$\therefore \frac{4}{3}\text{hv}_0 - \text{h}v_0 = \frac{1}{2}\text{mv}'^2$

$\therefore \frac{\text{v}'^2}{\text{v}^2} = \frac{\frac{4}{3}\text{v}-\text{v}_0}{\text{v}-\text{v}_0} \qquad \therefore \text{v}' = \text{v}\sqrt{\frac{\frac{4}{3}\text{v}-\text{v}_0}{\text{v}-\text{v}_0}}$

$\therefore \text{v}' > \text{v}\sqrt{\frac{4}{3}}$

Q.12 Answer:



Solution:

In X-ray tube, $\lambda_{\min} = \frac{hc}{eV}$

$$\text{In } \lambda_{\min} = \text{In}\left(\frac{hc}{e}\right) - \text{In } V$$

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Clearly, $\log \lambda_{\min}$ **versus** $\log V$ **graph**

slope is negative hence option (c) correctly depicts.

Q.13 Answer:

$\frac{P}{2C}$

Solution:

energy 1 emitted by the source Per unit time $= P$

$\text{Area } = 4\pi r^2$

energy falling on small area ot $dA = \frac{P}{4\pi R^2} \cdot dA$

$dp = \frac{dE}{c} = \frac{pdA}{4\pi R^2 c}$

Change in momentum per unit time $= \frac{PdA}{4\pi R^2 c} - \left(\frac{-PdA}{4\pi R^2 c}\right)$

$= \frac{2Pdn}{4\pi R^2 c}$

force on area $dA = dF = \frac{2PdA}{4\pi R^2 c}$

force along $0x = dF\cos\theta = \frac{2pdA\cos\theta}{4\pi R^2 C}$

$$dF\cos\theta = \frac{2P\left(dA\cos\theta^{\top}\right.}{4\pi R^2 c}$$

$$dF\cos\theta = \frac{2P}{4\pi R^2 c}(\text{ projection of } dA)$$

$$F = \frac{2P}{4\pi R^2 C}\int \text{ projection otd } A$$

$$= \frac{2P}{4\pi R^2 C} \times \pi R^2 = \frac{P}{2C}$$

Q.14 Answer:

is greater than $3E_k$

Solution:

We write Einstein's equation for the photoelectric effect, $\mathrm{h}f - W = E_K$**...(i)** $3\ \mathrm{h}f - W = E'_K$**...(ii)**

From (i) and (ii), we get, $E'_K - 3E_K = 2W > 0$

Q.15 Answer:

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Solution:

Number of emitted electrons $\mathrm{N_E} \propto$ Intensity

$$\propto \frac{1}{(\text{ Distance })^2}$$

Therefore, as distance is doubled, N_E decreases by $(1/4)$ times.

Q.16 Answer:

$1 : 2$

Solution:

$\mathrm{E} = \mathrm{W_0} + \mathrm{K_{max}}$
$\Rightarrow \mathrm{hf} = \mathrm{W_A} + \mathrm{K_A}$
and $2\mathrm{hf} = \mathrm{W_B} + \mathrm{K_B} = 2\ \mathrm{W_A} + \mathrm{K_B}$
$\left(\because \frac{\mathrm{W_A}}{\mathrm{W_B}} = \frac{1}{2}\right)$
Dividing equation (i) by (ii) $\frac{1}{2} = \frac{\mathrm{W_A}+\mathrm{K_A}}{2\ \mathrm{W_A}+\mathrm{K_B}} \Rightarrow \frac{\mathrm{K_A}}{\mathrm{K_B}} = \frac{1}{2}$

Q.17 Answer:

$2.2\mathrm{eV}$

Solution:

Energy required to excite electron to first excited state is ΔE=E2−E1 =−3.4−(−13.6)=10.2eV.
Work Function = ΔE - KE = 10.2 - 8 = 2.2 eV

Q.18 Answer:

0.57

Solution:

Use $\mathrm{n} = \frac{\mathrm{I}}{\mathrm{h\nu}} = \frac{\mathrm{I}\lambda}{\mathrm{hc}}$
$\therefore \frac{\mathrm{n_{UV}}}{\mathrm{n_{IR}}} = \frac{\lambda_{\mathrm{UV}}}{\lambda_{\mathrm{IR}}} = \frac{400}{700} = \frac{4}{7} = 0.57$

Q.19 Answer:

$\mathrm{V_0}$

Solution:

Since, stopping potential is independent of distance hence new stopping potential will remain unchanged i.e., new stopping potential $= V_0$.

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Q.20 Answer:

$N = P\lambda d^2 t/hcl^2$

Solution:

$E = \frac{hc}{\lambda}$

Number of photons emitted is

$$\frac{Pt}{\left(\frac{hc}{\lambda}\right)} = n_0 \quad n_0 = \frac{P\lambda t}{hc}$$

Since the radiation is spherically symmetric, so total number of photons entring the sensor is n_0 times the ratio of aperture area to the area of a sphere of radius ℓ.

$$N = n_0 \frac{\pi(2\,d)^2}{4\pi\ell^2} = \frac{P\lambda t}{hc}\frac{d^2}{\ell^2}$$

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